



CONCEPT NOTE

Humanitarian Attention-Gap Intelligence

From fragmented evidence to explainable rankings and timely outreach.

LIVE PROTOTYPE

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EXECUTIVE SUMMARY

Lumen identifies humanitarian crises where human need is high but public and media attention is disproportionately low. It combines displacement, appeal, funding, disaster and coverage signals; calculates a transparent Attention-Gap Score; and converts the evidence into grounded briefs for journalists, NGOs and donors.

The deployed platform creates one repeatable flow from public data to ranking, explanation and action through a web dashboard, Telegram bot and email delivery.

PROJECT AT A GLANCE

26

initial crises

monitored as one
comparable cohort

Daily

automated flow

ingest, score, brief
and deliver

3

delivery surfaces

dashboard, Telegram
and email

Live

production system

HTTPS, containers
and scheduled n8n

CORE PROPOSITION

**Reveal where need is high but attention is low—
then turn the evidence into a
ready-to-share brief.**

One clear ranking. Traceable facts. Immediate outreach.



THE NEED

A visibility problem with real consequences

Humanitarian attention does not always follow humanitarian need. Severe displacement, large response appeals and major funding gaps may coexist with low media visibility. The evidence exists, but it is fragmented across specialist systems and difficult to turn quickly into a defensible priority or communication action.

1 Fragmented evidence

Teams manually reconcile humanitarian, funding and coverage datasets.

2 Delayed discovery

Slow comparison delays reporting, advocacy and funding attention.

3 Weak traceability

Generic alerts and AI summaries rarely expose a reproducible evidence trail.

PROJECT GOAL

Improve the visibility of under-reported humanitarian crises

Transform trusted public data into explainable prioritization and timely outreach.

OBJECTIVES

- Harmonize crisis, displacement, funding and coverage signals.
- Calculate an auditable, versioned Attention-Gap Score.
- Present rankings, trends, evidence and briefs through a public dashboard.

- Generate audience-specific briefs without invented statistics.
- Deliver priority findings through Telegram and email every day.
- Build historical evidence for research, evaluation and model improvement.

INTENDED USERS

Journalists & editors

News and story discovery

NGO / UN analysts

Advocacy and monitoring

Donor & policy teams

Priority and grant decisions

University labs

Research and teaching



EVIDENCE TO ACTION

How Lumen works

A daily automated pipeline connects public evidence, transparent scoring and direct outreach.

1 COLLECT

n8n triggers source-specific ingestion once per day.

2 STORE

PostgreSQL preserves raw readings before calculation.

3 SCORE

A tested engine normalizes available cohort signals.

4 RANK

Need versus coverage, adjusted for underfunding.

5 EXPLAIN

Dashboard shows trends, evidence and grounded briefs.

6 ACT

Priority briefs reach users through Telegram and email.

SCORING METHOD

Transparent, cohort-relative prioritization

- Log-transform inputs so very large displacement values do not flatten the cohort.
- Normalize available signals to 0–1 across crises scored in the same run.
- Estimate need from displacement, appeal size and active-disaster count.
- Compare need with media coverage, then amplify and add the funding-gap signal.

DATA SOURCES

UNHCR Displaced people

OCHA FTS Appeals and funding gap

GDELT Media coverage volume

ReliefWeb Active disaster count

ReliefWeb activates after appname approval.

GROUNDING SAFEGUARDS

- Scores store algorithm version and inputs.
- Every brief references its exact score row.
- Numeric facts are rendered deterministically.
- Groq failure triggers a grounded fallback.

No source failure or AI outage is allowed to erase the evidence trail.



IMPLEMENTATION STATUS

A working end-to-end production prototype

The platform is deployed over HTTPS and the complete data-to-notification loop has been tested.

- ✓ PostgreSQL schema, migrations and idempotent evidence storage
- ✓ Tested Attention-Gap scoring engine and Express API
- ✓ Live UNHCR and OCHA FTS ingestion with resilient source adapters
- ✓ Next.js dashboard with rankings, trends, evidence and failure states
- ✓ Grounded brief generation plus live-tested Telegram and email delivery
- ✓ Daily n8n automation, CI/CD, containers and public production deployment

CURRENT LIMITATION

ReliefWeb needs a pre-approved appname. Until approval, that optional signal is skipped; other sources continue producing rankings. GDELT rate limits are handled with pacing and retry.

WHY IT IS DIFFERENT

Transparent priority

Need-versus-attention logic instead of an opaque score.

Replayable evidence

Raw readings and versioned inputs support audit and reuse.

Built for action

One workflow joins discovery, explanation and delivery.

EXPECTED OUTCOMES

- Faster discovery of overlooked crises
- Lower manual research effort
- More traceable public-interest briefs
- Greater visibility for underfunded crises

KEY INDICATORS

- Source freshness and success rate
- Crises successfully scored
- Brief and delivery success rate
- Users, shares and downstream actions

OPERATIONAL READINESS

Risks are known, visible and manageable

RISK	MITIGATION
API failure or rate limiting	Timeouts, retry/backoff, pacing and partial-source scoring
Missing or uneven data	Drop missing inputs, re-normalize weights, expose evidence
Invented AI statistics	Deterministic facts, score linkage, validation and fallback
Score drift across cohorts	Retain raw inputs and versions for future recalculation
Secret exposure	Environment-only credentials, HTTPS and restricted routes
Cost after cloud credits	Spending alerts, usage review and a smaller pilot deployment

SUSTAINABILITY MODEL

Keep public value free; fund advanced institutional use

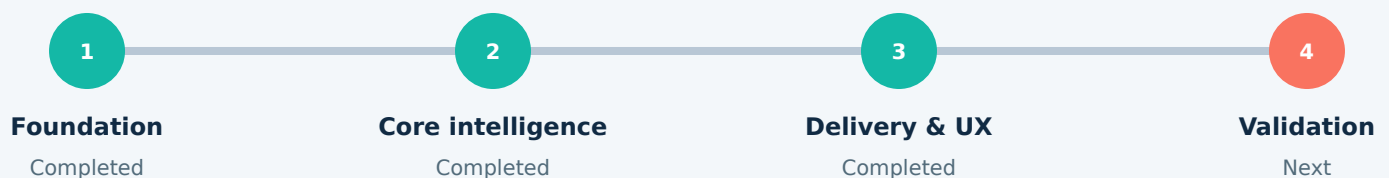
- Free public dashboard for visibility, trust and discovery
- Team subscriptions for custom alerts, exports and history
- Institutional API access for newsrooms, NGOs and donors
- Sponsored monitoring, research grants and philanthropy

CURRENT BASE COST

≈ **\$24**
per month

Production server
Variable provider, domain
and monitoring costs extra

WORK PLAN



NEXT PRIORITY

Validate the deployed prototype with journalists, humanitarian analysts and donor teams; activate ReliefWeb after approval; calibrate the scoring model; and document real-world use.